

Walkthrough – Create, Compile and run CAMkES projects

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Get CAMkES and install sources

The official tutorials for CAMkES will be find in <https://docs.sel4.systems> .

You can install sources with the following instructions. Make sure, that **repo** is already installed.

```
mkdir <DIR_NAME>
cd <DIR_NAME>
repo init -u https://github.com/sel4/camkes-manifest.git
repo sync
```

CAMkES projects should be compiled in sel4 environment. So make sure, that sel4 project is installed and will be compiled with the target **user_camkes** .
NOTE: Placeholder **<DIR_NAME>** is representative for fully qualified path name

Create, Compile and run a CAMkES project

Test projects are located in path **<DIR_PATH>/projects/camkes/apps** from your CAMkES installation. If you want to compile any projects follow these steps.

HINT: Before you will run the docker container in QEMU, change to the directory level in which you can find your installation.

1. copy your CAMkES project to path **<DIR_PATH>/projects/camkes/apps** or create at this a directory
2. run the docker container in QEMU with statement: **make -C <DIR_PATH>sel4-CAMkES-L4v-dockerfiles user_camkes HOST_DIR=\$(pwd)**
3. create a directory **build** with subdirectories according your project name in the CAMkES installation directory
4. change to subdirectory **<DIR_PATH>/build/<PROJECT_NAME>**
5. invoke the script **init-build.sh** to copy the sources to your project directory (you can find it in your installation directory on the entry level)

- tell the script the project, which want to be compile with:

```
-DCAMKES_APP=<PROJECT_NAME>
```

HINT: The **<PROJECT_NAME>** has the same identifier like your created project folder. If you deal with other identifiers, like the project directories identifiers, you become an error message.

- tell the script the platform, which want to be compile with:

```
-DPLATFORM=<PLATFORM>
```

HINT: The current **<PLATFORM>** is **zynq7000**

- HINT: The current status of sel4 forces you to specify an additional parameter when calling the script: **-DENABLE_LINT=OFF**. If you don't specify this parameter, currently you will getting an error message.
6. compile the project with **ninja** in you current directory
 7. run the project with invoke the script **simulate** in your current directory

Summary about all steps:

```
cp -r <SOURCE_PATH> <DIR_PATH>/projects/camkes/apps
mkdir -p <DIR_PATH>/build/ <PROJECT_NAME>
make -C <DIR_PATH>sel4-CAMkES-L4v-dockerfiles user_camkes HOST_DIR=$(pwd)
cd <DIR_PATH>/build/<PROJECT_NAME>
../../init-build.sh -DPLATFORM=<PLATFORM> -DENABLE_LINT=OFF -DCAMKES_APP=<PROJECT_NAME>
ninja
./simulate
```

NOTE: It doesn't purposeful to try to change the project directory in your installed CAMkES directory. You only will be kill some time.

Now you can create own projects or comprehend the CAMkES projects.

You can download and try the attached projects to learn CAMkES.